



EAA

Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning

Hongfei Du, Sidi Lu, Gang Zhou & Ye Gao

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Department of Computer Science, William & Mary

Making audio LLMs emotion-aware

Why this paper matters



The first paper in this series whose whole purpose is emotion

0.687

accuracy on MELD — the best
audio LLM result on this set

+11.4

points over the strongest
prior audio LLM baseline

2

encoders, fused by attention
in both directions

1

H100, LoRA rank 2 —
a small-lab result

● The claim

Audio language models handle speech well and emotion badly for two specific reasons: they use a single encoder, or fuse two encoders by simple concatenation; and they judge each utterance in isolation when emotion evolves across a dialogue. Fix the fusion with bidirectional cross-attention and fix the isolation by putting the preceding utterance into the prompt.

● Comments

Every model we have studied appears in this paper's baseline table, and every one of them is beaten by a wide margin: Pengi at 0.289, SALMONN at 0.331, Qwen-Audio at 0.557. EAA reaches the level of dedicated multimodal conversation models that also use video.

1

Where the series left emotion

Six sessions of evidence that something is missing



The running finding



What we have established across the term

● The corpora are noisy

IEMOCAP's raters reach a Fleiss kappa of 0.27; CREMA-D's two annotators agree on the category 60.7% of the time. Every emotion result sits against a ceiling set by that disagreement.

● The benchmarks under-weight it

SUPERB gives paralinguistics one task out of fifteen, and that task is IEMOCAP four-class. The emotion column is the flattest in the whole table.

● The objectives discard it

wav2vec 2.0's quantisation bottleneck abstracts away from speaker characteristics by design, and ASR fine-tuning measurably damages emotional information — 58.5 to 44.6 on the context encoder.

● The audio LLMs are thin on it

Pengi at chance zero-shot on CREMA-D. SALMONN at 0.69 against an 0.81 specialist, trained on five hours of IEMOCAP in a 4,400-hour mix. Qwen-Audio: one weighted accuracy on MELD from under 1,000 hours.

The two challenges the paper names



Both are claims about existing audio LLMs

● Challenge 1 — feature integration

Most models rely solely on a single audio encoder. Others extract semantic and acoustic features separately and merely concatenate them, failing to capture the intricate interactions between the two feature types.

The paper's survey is specific: Pengi uses CLAP alone; Qwen-Audio and OSUM use Whisper alone; SALMONN splices Whisper and BEATs before a Q-Former; WavLLM fuses Whisper and WavLM with 1D convolution and adapters.

● Challenge 2 — context

Emotions in dialogue evolve progressively within context, making it difficult to perceive the overall emotional state from isolated audio segments.

The paper's example: the utterance 'What?' can be angry, neutral or surprised depending on whether it follows 'Did you just insult me?' or a piece of shocking news.

Most existing audio LLMs do not incorporate contextual information at all.

● Both are fair descriptions of what we read

SALMONN concatenates Whisper and BEATs features frame by frame along the feature dimension — exactly the splicing the paper criticises. Qwen-Audio uses one encoder. None of the four audio LLMs we studied takes conversational context as input.

So the diagnosis is accurate. The question the rest of the talk asks is whether the two proposed fixes are what produce the gain.

Why fusion, specifically



Emotion is carried by two channels that trade off

● When acoustics dominate

High pitch for anger, slow speech for sadness. The words may be neutral or even contradict the delivery.

Here the model should weight acoustic features more heavily.

● When words dominate

Explicit emotion words — 'happy', 'angry' — carry the label directly, and the delivery may be flat.

Here the model should prioritise linguistic features.

● The design goal, stated in the introduction

A mechanism that dynamically adjusts feature importance based on context — so the weighting between the two channels is learned per example rather than fixed.

Compare with the CREMA-D result from the corpora session: some emotions are carried by one channel and some by another, and the optimal weighting is emotion-dependent. That finding is the empirical case for exactly this design.

2

Dual cross-attention fusion

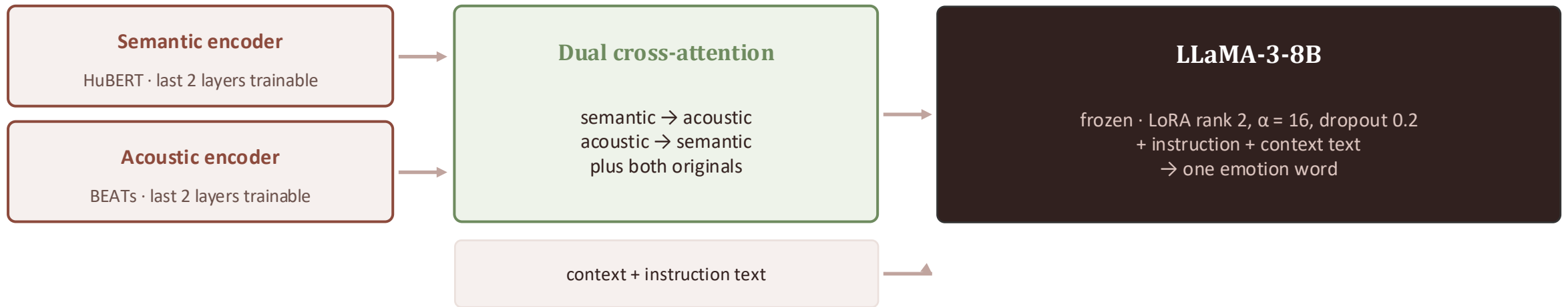
The first of two contributions



The architecture



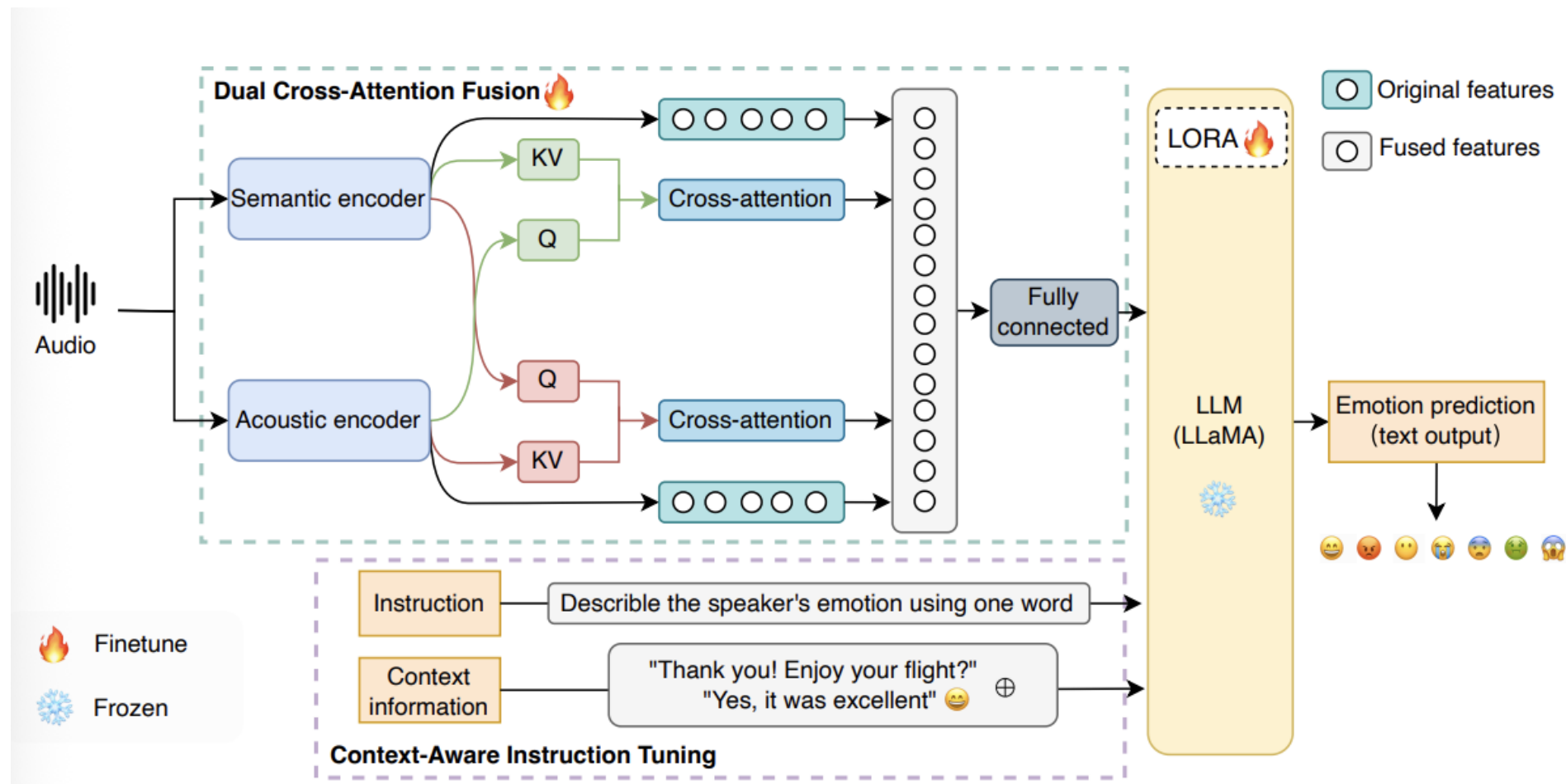
Two encoders, a fusion block, a frozen LLaMA with LoRA



- Both encoders are frozen except their last two layers — a middle position between fully frozen and fully trained.
- The fused representation is projected into LLaMA's hidden space and concatenated with the instruction and context text.
- The instruction is fixed: 'Describe the speaker's emotion in one word.' The output is a word, not a class index.
- Note what is not here: no Q-Former, no adapter stack, no separate classification head. The LLM produces the label as text.

The architecture

Two encoders, a fusion block, a frozen LLaMA with LoRA



The two encoders



Both are models we have read in full

● HuBERT as the semantic encoder

The paper's justification: HuBERT leverages pseudo-labels generated from spectral features or its own learned representations, which have been shown to contain rich semantic information.

From our HuBERT session: masked prediction of offline k-means cluster targets, bootstrapped from MFCCs and refined twice. The paper cites HuBERTopic for the semantic claim.

● BEATs as the acoustic encoder

The paper's justification: BEATs captures fine-grained spectral and prosodic characteristics directly from the raw audio signal.

From our SALMONN session: BEATs is trained with iterative self-supervised learning and an acoustic tokeniser distilled from semantic knowledge — and SALMONN uses it for exactly the same non-speech-semantics role.

● Question

Is HuBERT a semantic encoder? The layer-wise analysis we read in the wav2vec 2.0 session says it depends entirely on which layer you take: shallow layers are acoustic, middle layers phonetic, upper-middle layers lexical, and the deepest layers drift back toward the input.

The paper does not say which layer it uses, and it fine-tunes the last two — which by that analysis are the least semantic ones in the stack.

Getting the two streams into the same space



Three preparatory steps before any attention happens

1 Project

Two trainable matrices map the HuBERT output S and the BEATs output A to a common dimension d : $\tilde{S} = W_s S + b_s$ and $\tilde{A} = W_a A + b_a$.

2 Normalise

Layer normalisation is applied to both projected streams to stabilise training — necessary because the two encoders were trained with different objectives and have different activation scales.

3 Pad

The two sequence lengths T_s and T_a may differ. The shorter sequence is zero-padded along the temporal axis to $T = \max(T_s, T_a)$, so attention operates over temporally aligned representations.

• The padding step

Zero-padding aligns the sequences in length but not in time. HuBERT and BEATs both output at 50 Hz, so the frame rates match — but any offset in windowing or downsampling would mean frame t in one stream and frame t in the other are not the same moment of audio.

Dual cross-attention



Each stream queries the other

● Semantic queries acoustic

$$\text{Att}_s = \text{Softmax}(Q_s K_a^T / \sqrt{d}) V_a$$

The semantic features ask: for each word-level position, which acoustic moments matter? This incorporates detailed acoustic characteristics into the linguistic representation.

● Acoustic queries semantic

$$\text{Att}_a = \text{Softmax}(Q_a K_s^T / \sqrt{d}) V_s$$

The acoustic features ask: for each prosodic moment, which words matter? This enriches the acoustic representation with linguistic information.

● What 'dual' buys over 'single'

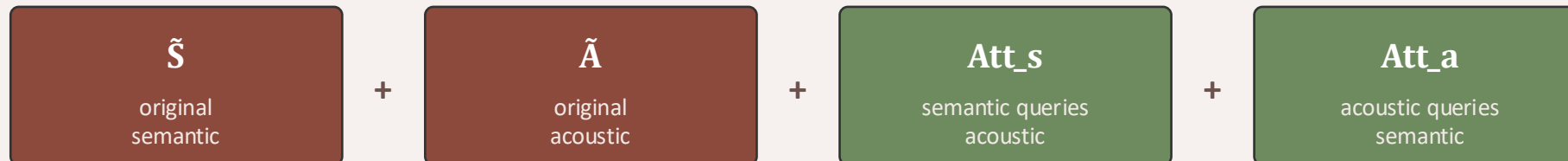
A single cross-attention is asymmetric: one stream is enriched and the other is only consumed. Running it both ways means each representation is conditioned on the other, and neither is subordinate.

The paper's stated motivation is adaptivity: when acoustic cues predominate the model learns to weight them more, and when explicit emotion words dominate it prioritises the linguistic side. Whether the learned attention actually behaves that way is not measured.

Why the originals are kept too



The fusion is four streams, not two



● The stated reason

To maintain feature integrity while enhancing fusion. Concatenating the originals preserves unique feature characteristics while ensuring comprehensive integration.

● Why this matters in practice

Attention output is a weighted average of value vectors. If a distinctive acoustic cue has no semantic anchor to attend from, it can be averaged away. Keeping the raw streams guarantees it still reaches the language model.

● A familiar pattern

This is a residual connection in all but name, and it is the same instinct behind SUPERB's weighted layer sum and WavLM's layer aggregation: do not force the downstream model to accept one transformed view when you can hand it several and let it choose.

3

Context-aware instruction tuning

The second contribution



Context-aware instruction tuning



One preceding utterance, concatenated

● The mechanism

For an audio sample x_t at time t , retrieve the most recent preceding utterance x_{t-1} within the same conversation and concatenate at sentence level:

$$C_t = x_{t-1} \oplus x_t$$

If no prior utterance exists, the current one is used alone.

● The instruction

'Describe the speaker's emotion in one word.'

The stated purpose is to make the output align with the expected emotion categories — a generative model asked for a single word is far more likely to produce something that maps cleanly onto one of the seven labels.

The final input to LLaMA is the fused audio features, the instruction, and the context.

● A question the paper leaves implicit

Is the context audio, or text? The method section says 'preceding audio utterances' and 'contextual representation'; the ablation table describes the settings in terms of whether text information about the audio is provided.

The distinction matters enormously for what the model is doing, and the ablation on the next slide is where it becomes decisive. Hold the question until then.

Implementation



A small-lab configuration

Component	Setting	Note
Audio	16 kHz, single channel, resampled as needed	Standard
Semantic encoder	HuBERT, all layers frozen except the last two	Partial adaptation
Acoustic encoder	BEATs, all layers frozen except the last two	Same treatment
Language model	LLaMA-3-8B with LoRA	Frozen backbone
LoRA	rank 2, $\alpha = 16$, dropout 0.2	Rank 2 is unusually low
Optimiser	AdamW, learning rate 5×10^{-6} , warm-up then cosine decay	Conservative rate
Batch size	2	Very small — no gradient accumulation reported
Hidden size	768	The fusion dimension
Hardware	One H100 GPU	Compare Qwen-Audio's 500k-step industrial run

No training duration, epoch count, or seed is reported, and results are single runs.

Part I in four sentences



Before the break

- 1 Across six sessions we established that audio language models handle emotion badly, for reasons ranging from noisy corpora to objectives that discard paralinguistic information.
- 2 EAA diagnoses two specific causes: single-encoder or concatenation-based fusion, and judging each utterance in isolation when emotion evolves across a dialogue.
- 3 The fusion fix is bidirectional cross-attention between a HuBERT semantic stream and a BEATs acoustic stream, with both original streams concatenated back in.
- 4 The context fix is to prepend the immediately preceding utterance, with a fixed instruction asking for the emotion in one word.

4

Results and ablations

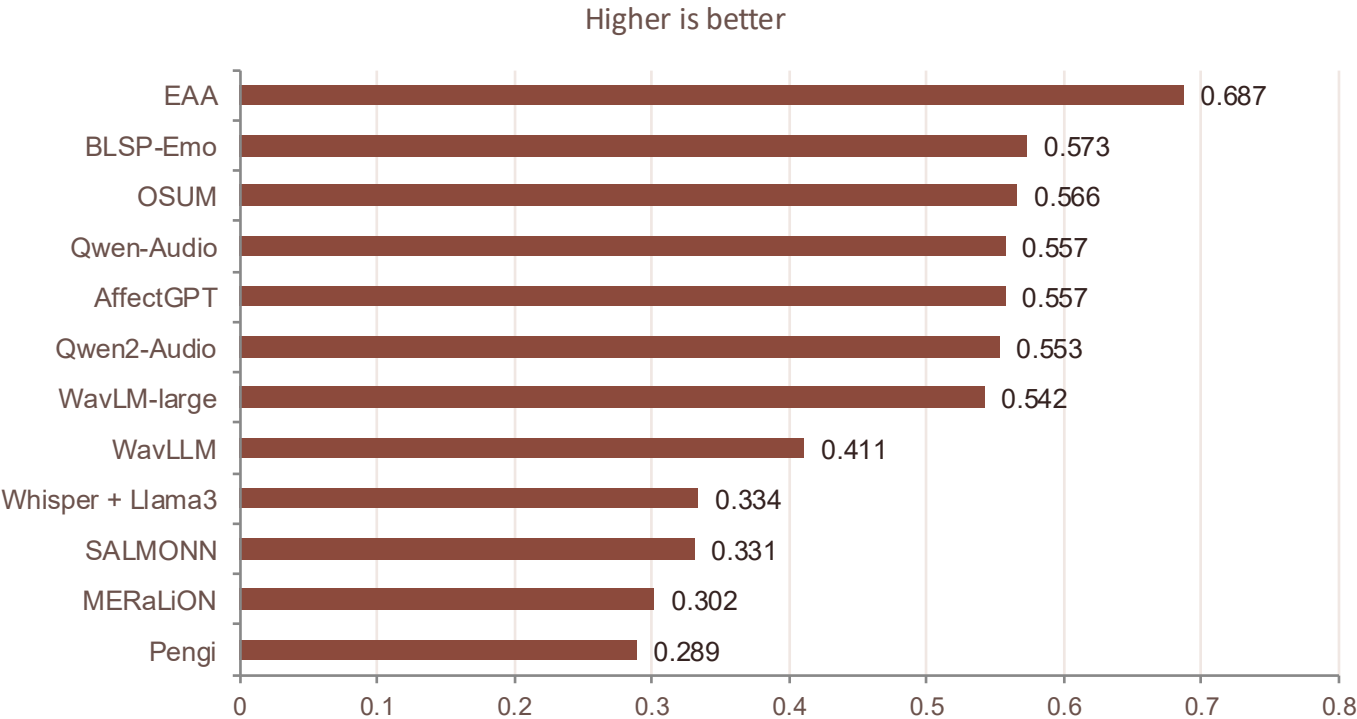
The table, and the three rows that reframe it



Against audio language models



MELD accuracy — every model in this table is one we have read or discussed



- EAA at 0.687 against the best prior audio LLM, BLSP-Emo, at 0.573 — the 11.4-point gain in the abstract.
- The ordering is informative: general-purpose models cluster at 0.55 to 0.57, and the models that predate them sit near 0.30.
- Pengi at 0.289 on seven classes is barely above the 0.14 chance level — the same picture as its 0.185 on CREMA-D.
- WavLM-large at 0.542 is a frozen self-supervised encoder with a classification head, and it beats four audio LLMs.

The fourth bullet is the series' recurring embarrassment: a representation model with a linear head outperforming models built on eight-billion-parameter LLMs.

The baselines



Four things the superscripts tell you

● Three provenances

The table marks results cited from original papers, results reported by AudioBench, results reported by Qwen-Audio, and results the authors ran themselves. Properly disclosed, and it means decoding, prompting and label-mapping differ across rows.

● Generative versus classification

Most baselines generate a word that must then be matched to one of seven labels. How that matching is done is not stated for EAA or for the cited baselines — and from the Pengi session we know text-matching choices are worth several points.

● Accuracy only

One number per model. MELD is heavily imbalanced toward neutral, and from the CLEP-DG session we know a weighted accuracy near 50 can accompany an unweighted accuracy near 23.

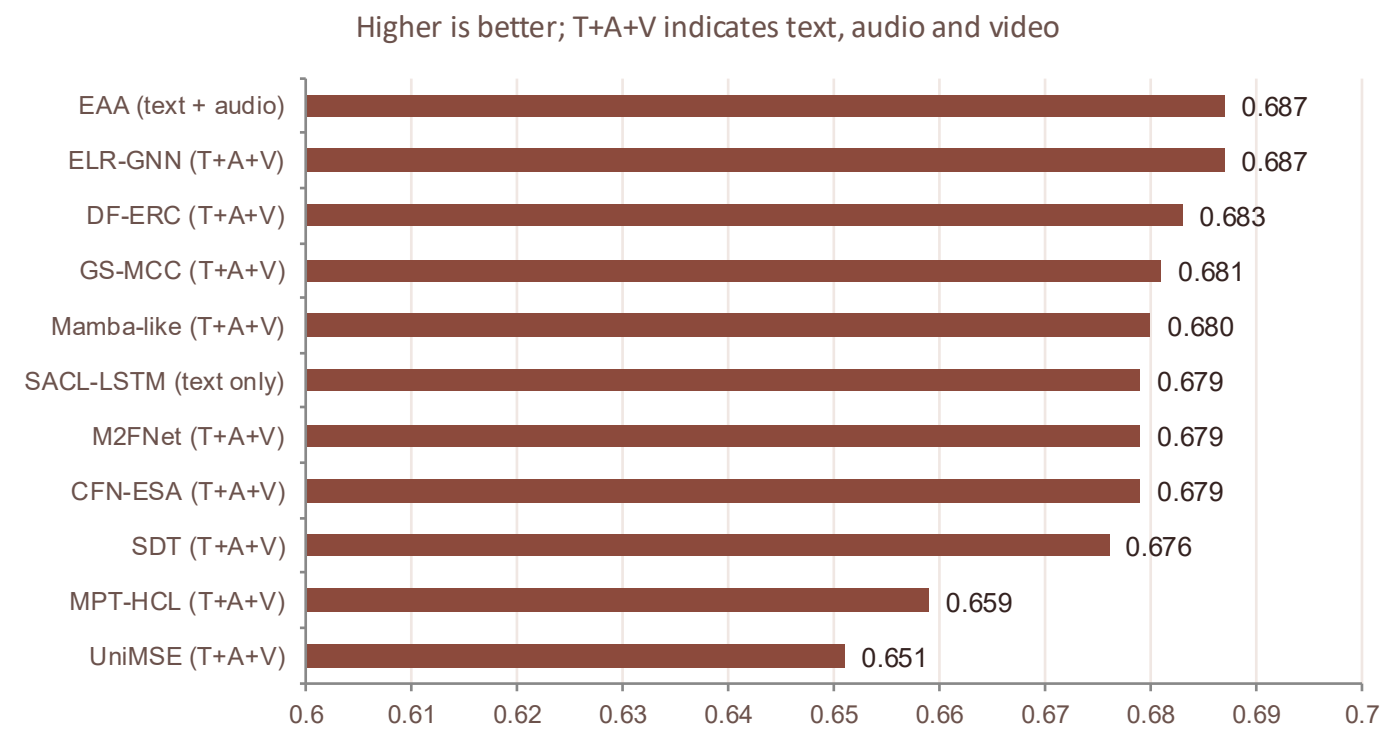
● The comparison that is controlled

None of them, strictly. The internally controlled evidence is the ablation table, which holds the model fixed and varies one factor — and that is the next slide.

Against purpose-built conversation models



Most of these use video as well



- EAA matches ELR-GNN, the strongest specialist, which also uses video.
- The paper's own reading: most audio LLMs lag traditional classifiers because classification picks from seven categories while generation must select an emotion word from a much larger vocabulary.
- The framing is that EAA closes the gap between audio LLMs and purpose-built classifiers — which is a fair and appropriately modest claim.
- But note the axis: this chart starts at 0.60. The spread across eleven systems is under four points.

Reaching a specialist's accuracy with one fewer modality is the strongest version of the claim, and it is the one the paper makes.

One observation



SACL-LSTM uses text and nothing else

0.679

SACL-LSTM — text only,
no audio, no video

0.687

EAA — text and audio,
dual cross-attention fusion

- **Eight tenths of a point for the entire audio pathway**

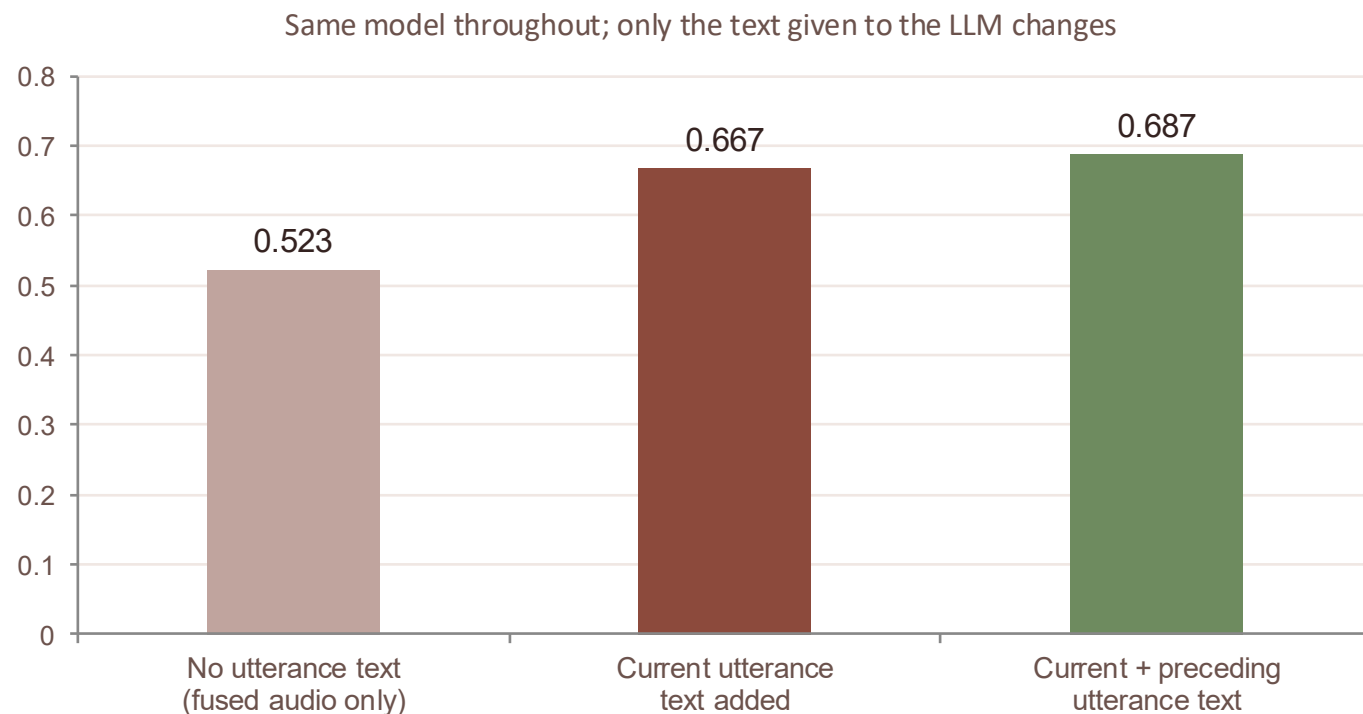
A supervised contrastive LSTM over transcripts alone reaches 0.679 on MELD. EAA, with two self-supervised audio encoders, bidirectional cross-attention and an eight-billion-parameter language model, reaches 0.687.

This is not an argument against the paper — the same gap applies to every multimodal system in the table, several of which add video and gain less than a point. It is an argument about MELD, and about how much headroom the audio channel has on this dataset at all.

The ablation that reframes the paper



Three rows, and the gap between the first two



- The paper's own description of row one: neither the current utterance nor the preceding sentence is provided — there is no additional text information about the input audio.
- So row one is the full audio pipeline — both encoders, dual cross-attention, LLaMA — with no transcript. It reaches 0.523.
- Adding the current utterance's text is worth 14.4 points. Adding the preceding utterance's text is worth a further 2.0.
- The named contribution — context — accounts for 2.0 of the 16.4 points in this table.

This is the most informative table in the paper, and it says the dominant factor is the transcript of the current utterance.

What that implies



Three readings, and how to tell them apart

● The transcript is doing the work

MELD's emotion is largely lexical — Friends dialogue where the script carries the affect. The text-only SACL-LSTM at 0.679 supports this. Under this reading, EAA is a strong text model with an audio assist.

● The audio pathway is under-trained

0.523 from two frozen-except-last-two-layers encoders, a fusion block, and LoRA rank 2 may simply reflect a small adaptation budget rather than a ceiling on what audio can contribute.

● The fusion needs the text to be useful

The cross-attention output may only become discriminative once the LLM has linguistic context to anchor it. Under this reading the components are complementary and the decomposition is not additive.

● The experiment that separates them

Run LLaMA with the same LoRA setup on transcripts alone, no audio at all. If that reaches the high sixties, reading one is right. If it lands well below 0.667, the audio pathway is contributing more than the ablation's first row suggests and reading three is in play.

It is one training run and it is the single most informative number missing from the paper.

The attention ablation



Four configurations, reported as a figure

● Self-attention only

Acoustic and semantic features each undergo self-attention with no interaction. The paper reports this still achieves relatively good performance, but is not as effective as dual cross-attention.

● Single: acoustic queries semantic

Acoustic features as query, semantic as key and value. Reported as the better of the two single-direction configurations.

● Single: semantic queries acoustic

The reverse. Reported as weaker — cross-attention with acoustic features attending to semantic representations captures more useful information than the opposite direction.

● Dual (the proposed method)

Both directions, plus the two original streams concatenated. The best of the four.

The ordering is reported; the values are in a figure rather than a table, so no numbers are quoted here.

5

What is established, and what is next

An appraisal, and five experiments



Appraisal



What the design buys, and the open threads

● Strengths

An accurate diagnosis: we can verify the survey of fusion strategies against our own reading of Pengi, SALMONN, Qwen-Audio and WavLLM.

A clean, simple mechanism. Bidirectional cross-attention with the originals preserved is easy to describe, easy to implement, and motivated by a real property of emotional speech.

The best audio-LLM result on MELD by a wide margin, and parity with a video-using specialist.

An honest ablation. The audio-only row is not flattering, and reporting it is what makes the decomposition visible at all.

Achieved on one GPU with LoRA rank 2 — an academic-scale result in a literature dominated by industrial training runs.

● Open threads

No text-only control, so the contribution of the audio pathway cannot be bounded from above.

One dataset. MELD only, where a text-only LSTM reaches 0.679 and every extra modality adds under a point.

Accuracy only, on a corpus known to be imbalanced — no unweighted accuracy, F1, or per-class breakdown.

The label-mapping step from generated word to one of seven classes is not described.

The adaptivity claim about the attention is motivating rather than measured — no attention analysis is reported.

No seeds, epochs, or variance; single runs throughout.

Five experiments that would strengthen it



Roughly in order of value per unit of effort

- 1 The text-only control**

LLaMA-3-8B with the same LoRA setup on transcripts alone. Together with the existing audio-only row this brackets both channels and settles the three readings from the previous section. One run.
- 2 A second corpus where audio matters**

MELD is scripted television where the text carries the affect. IEMOCAP or MSP-Podcast would test whether dual cross-attention helps when the words are less informative — and MSP-Podcast ships speaker-independent splits and a non-retrieved control partition.
- 3 Report unweighted accuracy**

MELD is imbalanced toward neutral. A weighted accuracy of 0.687 is compatible with weak performance on the rare emotions, and the per-class picture is what a downstream user needs.
- 4 Inspect the attention**

The adaptivity story predicts that acoustic weight rises on prosody-carried emotions and falls on lexically explicit ones. Attention maps are free to extract and would turn the motivation into a finding.
- 5 Sweep the HuBERT layer**

The layer-wise analysis says the deepest HuBERT layers are the least semantic. A layer sweep, or a learned weighted sum as SUPERB and WavLM use, is cheap and may move the audio-only number.

6

Synthesis

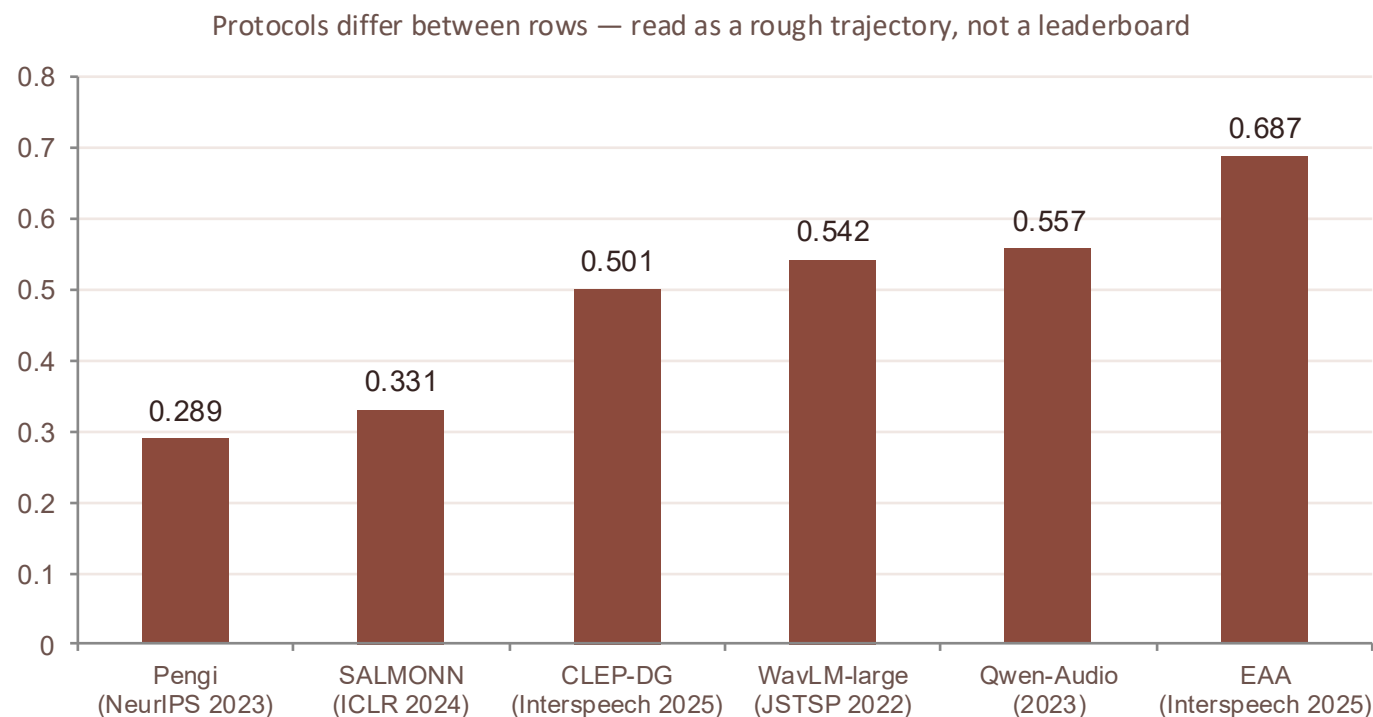
Where this leaves the series



MELD across the series



Every model we have read that reports this dataset



- CLEP-DG's figure is weighted accuracy under its own protocol and is included for context, not as a like-for-like comparison.
- The trajectory is real: from barely above chance in 2023 to the level of purpose-built conversation models in 2025.
- But the EAA bar includes the transcript, and the ablation tells us 0.523 of it is the audio pathway alone.
- Read that way, the audio-only trajectory across the series is much flatter than this chart suggests.

The honest summary of the term: emotion recognition from audio language models has improved, and a large part of the visible improvement comes from giving the models the words.

What EAA adds to the picture



Three contributions that survive the appraisal

● A design that takes emotion as the goal

Every other model in the series treats emotion as one task among thirty. This is the first where the architecture is chosen for it — two encoders picked for complementary roles, fused adaptively.

That reframing is worth something independent of the numbers.

● Evidence that context matters

Two points from one preceding utterance is small but real, and no other audio LLM in the series takes conversational context at all.

It also suggests an obvious extension nobody has run: how much does a longer context window buy?

● A measurement of the channel split

Because the authors reported the audio-only row, we now have a number for how much of MELD emotion the audio pathway carries on its own: 0.523 against 0.687.

That number is more useful to the field than the headline.

● The framing

The series has shown repeatedly that emotion is the task where audio language models are weakest. This paper is the first to treat that as the problem to solve rather than a column in a table — and its most valuable output may be the ablation that shows how far there still is to go.

Take-aways



Five things to carry out of the series

- 1 Bidirectional fusion beats concatenation**

Each stream conditioned on the other, with both originals preserved. Simple, cheap, and better than splicing — the mechanism is the paper's durable contribution.
- 2 Always ask what the text is doing**

When a multimodal model has access to transcripts, the single most informative control is the text-only baseline. Without it, the modality contributions cannot be bounded.
- 3 Choose the dataset to fit the question**

On MELD a text-only LSTM reaches 0.679 and every extra modality adds under a point. A corpus where the words carry the emotion is a weak test of audio emotion.
- 4 Report the unflattering row**

The audio-only ablation is what makes this paper interpretable. A result you can decompose is worth more than a result you cannot.
- 5 Emotion is still the open problem**

Six sessions, four audio LLMs, two benchmark papers and four corpora, and the honest audio-only number on conversational emotion is in the low fifties.



Questions for discussion

- 1 The audio-only row reaches 0.523 and the full system 0.687. What would you predict a text-only control returns — and does your prediction change what you think the dual cross-attention is contributing?
- 2 On MELD a text-only LSTM reaches 0.679 and seven systems using text, audio and video span 0.651 to 0.687. Is MELD a good testbed for audio emotion at all, and what would you replace it with?
- 3 The paper motivates dual cross-attention by saying the model should weight acoustics when prosody carries the emotion and words when it does not. How would you test whether the learned attention actually does that?
- 4 HuBERT is used as the semantic encoder with its last two layers fine-tuned — the layers the layer-wise analysis says are least semantic. Does the 'semantic encoder' framing survive, and what would you change?
- 5 Across six sessions the audio-only number on conversational emotion has barely moved. Is the bottleneck the models, the corpora, the labels, or the evaluation — and which would you attack first?